

FIFA_Data_Analysis

March 29, 2025

1 Import modules and load datasets

```
[5]: import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
```

```
[6]: players = pd.read_csv("WorldCupPlayers.csv")
matches = pd.read_csv("WorldCupMatches.csv")
world_cup = pd.read_csv("WorldCups.csv")
```

```
[7]: players.head(5)
```

```
[7]:
```

	RoundID	MatchID	Team	Initials	Coach Name	Line-up	Shirt Number	\
0	201	1096	FRA	CAUDRON	Raoul (FRA)	S	0	
1	201	1096	MEX	LUQUE	Juan (MEX)	S	0	
2	201	1096	FRA	CAUDRON	Raoul (FRA)	S	0	
3	201	1096	MEX	LUQUE	Juan (MEX)	S	0	
4	201	1096	FRA	CAUDRON	Raoul (FRA)	S	0	

	Player Name	Position	Event
0	Alex THEPOT	GK	NaN
1	Oscar BONFIGLIO	GK	NaN
2	Marcel LANGILLER	NaN	G40'
3	Juan CARRENO	NaN	G70'
4	Ernest LIBERATI	NaN	NaN

```
[8]: matches.head(3)
```

```
[8]:
```

	Year	Datetime	Stage	Stadium	City	\
0	1930.0	13 Jul 1930 - 15:00	Group 1	Pocitos	Montevideo	
1	1930.0	13 Jul 1930 - 15:00	Group 4	Parque Central	Montevideo	
2	1930.0	14 Jul 1930 - 12:45	Group 2	Parque Central	Montevideo	

	Home Team Name	Home Team Goals	Away Team Goals	Away Team Name	\
0	France	4.0	1.0	Mexico	

1	USA	3.0	0.0	Belgium
2	Yugoslavia	2.0	1.0	Brazil

	Win conditions	Attendance	Half-time Home Goals	Half-time Away Goals	\
0		4444.0	3.0	0.0	
1		18346.0	2.0	0.0	
2		24059.0	2.0	0.0	

	Referee	Assistant 1	Assistant 2	\
0	LOMBARDI Domingo (URU)	CRISTOPHE Henry (BEL)	REGO Gilberto (BRA)	
1	MACIAS Jose (ARG)	MATEUCCI Francisco (URU)	WARNKEN Alberto (CHI)	
2	TEJADA Anibal (URU)	VALLARINO Ricardo (URU)	BALWAY Thomas (FRA)	

	RoundID	MatchID	Home Team Initials	Away Team Initials
0	201.0	1096.0	FRA	MEX
1	201.0	1090.0	USA	BEL
2	201.0	1093.0	YUG	BRA

```
[9]: world_cup.head(3)
```

```
[9]:   Year Country Winner Runners-Up Third Fourth GoalsScored \
0  1930  Uruguay  Uruguay   Argentina    USA  Yugoslavia      70
1  1934   Italy   Italy Czechoslovakia Germany   Austria      70
2  1938   France   Italy      Hungary   Brazil    Sweden      84
```


	QualifiedTeams	MatchesPlayed	Attendance
0	13	18	590.549
1	16	17	363.000
2	15	18	375.700

```
[10]: print("Number of rows in players table = ",len(players))
```

```
Number of rows in players table = 37784
```

```
[11]: print("Number of rows in matches table = ",len(matches))
```

```
Number of rows in matches table = 4572
```

```
[12]: print("Number of rows in world_cup table = ",len(world_cup))
```

```
Number of rows in world_cup table = 20
```

2 Data Cleaning

2.1 Handle missing values for players table

```
[15]: players.isna().sum()
```

```
[15]: RoundID          0
      MatchID          0
      Team Initials    0
      Coach Name       0
      Line-up          0
      Shirt Number     0
      Player Name      0
      Position         33641
      Event            28715
      dtype: int64
```

```
[16]: players['Position'].ffill(inplace=True)
```

```
[17]: players['Position'].value_counts()
```

```
[17]: Position
C      25691
GK     10827
GKC     1266
Name: count, dtype: int64
```

```
[18]: players['Event'].fillna("No Event", inplace=True)
```

```
[19]: players.isna().sum()
```

```
[19]: RoundID          0
      MatchID          0
      Team Initials    0
      Coach Name       0
      Line-up          0
      Shirt Number     0
      Player Name      0
      Position         0
      Event            0
      dtype: int64
```

2.2 Handle missing values for matches table

```
[21]: matches.isna().sum()
```

```
[21]: Year            3720
      Datetime        3720
      Stage           3720
      Stadium         3720
      City            3720
      Home Team Name   3720
      Home Team Goals  3720
      Away Team Goals  3720
```

```

Away Team Name      3720
Win conditions      3720
Attendance          3722
Half-time Home Goals 3720
Half-time Away Goals 3720
Referee            3720
Assistant 1        3720
Assistant 2        3720
RoundID           3720
MatchID           3720
Home Team Initials 3720
Away Team Initials 3720
dtype: int64

```

```
[22]: matches.head(3)
```

```

[22]:      Year      Datetime      Stage      Stadium      City \
0  1930.0  13 Jul 1930 - 15:00  Group 1      Pocitos  Montevideo
1  1930.0  13 Jul 1930 - 15:00  Group 4  Parque Central  Montevideo
2  1930.0  14 Jul 1930 - 12:45  Group 2  Parque Central  Montevideo

      Home Team Name  Home Team Goals  Away Team Goals  Away Team Name \
0      France          4.0            1.0      Mexico
1      USA            3.0            0.0      Belgium
2  Yugoslavia          2.0            1.0      Brazil

      Win conditions  Attendance  Half-time Home Goals  Half-time Away Goals \
0                4444.0          3.0                0.0
1            18346.0          2.0                0.0
2            24059.0          2.0                0.0

      Referee      Assistant 1      Assistant 2 \
0  LOMBARDI Domingo (URU)  CRISTOPHE Henry (BEL)  REGO Gilberto (BRA)
1      MACIAS Jose (ARG)  MATEUCCI Francisco (URU)  WARNKEN Alberto (CHI)
2    TEJADA Anibal (URU)  VALLARINO Ricardo (URU)  BALWAY Thomas (FRA)

      RoundID  MatchID  Home Team Initials  Away Team Initials
0      201.0    1096.0          FRA          MEX
1      201.0    1090.0          USA          BEL
2      201.0    1093.0          YUG          BRA

```

```
[23]: matches.tail(5)
```

```

[23]:      Year  Datetime  Stage  Stadium  City  Home Team Name  Home Team Goals \
4567   NaN      NaN   NaN   NaN   NaN   NaN          NaN
4568   NaN      NaN   NaN   NaN   NaN   NaN          NaN
4569   NaN      NaN   NaN   NaN   NaN   NaN          NaN

```

4570	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4571	NaN	NaN	NaN	NaN	NaN	NaN	NaN

	Away Team Goals	Away Team Name	Win conditions	Attendance	\
4567	NaN	NaN	NaN	NaN	
4568	NaN	NaN	NaN	NaN	
4569	NaN	NaN	NaN	NaN	
4570	NaN	NaN	NaN	NaN	
4571	NaN	NaN	NaN	NaN	

	Half-time Home Goals	Half-time Away Goals	Referee	Assistant 1	\
4567	NaN	NaN	NaN	NaN	
4568	NaN	NaN	NaN	NaN	
4569	NaN	NaN	NaN	NaN	
4570	NaN	NaN	NaN	NaN	
4571	NaN	NaN	NaN	NaN	

	Assistant 2	RoundID	MatchID	Home Team	Initials	Away Team	Initials
4567	NaN	NaN	NaN		NaN		NaN
4568	NaN	NaN	NaN		NaN		NaN
4569	NaN	NaN	NaN		NaN		NaN
4570	NaN	NaN	NaN		NaN		NaN
4571	NaN	NaN	NaN		NaN		NaN

```
[24]: matches.dropna(subset=['Year'],inplace=True)
```

```
[25]: matches.tail(3)
```

```
[25]:
```

	Year	Datetime	Stage	\
849	2014.0	09 Jul 2014 - 17:00	Semi-finals	
850	2014.0	12 Jul 2014 - 17:00	Play-off for third place	
851	2014.0	13 Jul 2014 - 16:00	Final	

	Stadium	City	Home Team Name	Home Team Goals	\
849	Arena de Sao Paulo	Sao Paulo	Netherlands	0.0	
850	Estadio Nacional	Brasilia	Brazil	0.0	
851	Estadio do Maracana	Rio De Janeiro	Germany	1.0	

	Away Team Goals	Away Team Name	Win conditions	\
849	0.0	Argentina	Argentina win on penalties (2 - 4)	
850	3.0	Netherlands		
851	0.0	Argentina	Germany win after extra time	

	Attendance	Half-time Home Goals	Half-time Away Goals	\
849	63267.0	0.0	0.0	
850	68034.0	0.0	2.0	
851	74738.0	0.0	0.0	

		Referee	Assistant 1	Assistant 2 \
849	Cneyt	AKIR (TUR)	DURAN Bahattin (TUR)	ONGUN Tarik (TUR)
850	HAIMOUDI Djamel	(ALG)	ACHIK Redouane (MAR)	ETCHIALI Abdelhak (ALG)
851	Nicola RIZZOLI	(ITA)	Renato FAVERANI (ITA)	Andrea STEFANI (ITA)

	RoundID	MatchID	Home Team Initials	Away Team Initials
849	255955.0	300186490.0	NED	ARG
850	255957.0	300186502.0	BRA	NED
851	255959.0	300186501.0	GER	ARG

2.3 Preprocessing matches table

```
[27]: matches['Home Team Name'].value_counts()
```

```
[27]: Home Team Name
Brazil      82
Italy       57
Argentina   54
Germany FR  43
England     35
..
Wales       1
Norway      1
rn">United Arab Emirates  1
Haiti       1
rn">Bosnia and Herzegovina 1
Name: count, Length: 78, dtype: int64
```

```
[28]: matches['Home Team Name'] = matches["Home Team Name"].str.
      ↪replace(r'rn">', '', regex=True)
```

```
[29]: matches['Home Team Name'].value_counts().head(10)
```

```
[29]: Home Team Name
Brazil      82
Italy       57
Argentina   54
Germany FR  43
England     35
Germany     34
Netherlands 32
France      31
Spain       30
Uruguay     28
Name: count, dtype: int64
```

3 Most Number of World Cup Winning Title

```
[31]: world_cup.head(2)
```

```
[31]:   Year Country Winner Runners-Up Third Fourth GoalsScored \
0  1930  Uruguay  Uruguay    Argentina    USA  Yugoslavia      70
1  1934   Italy   Italy Czechoslovakia Germany   Austria      70

   QualifiedTeams  MatchesPlayed  Attendance
0                13             18    590.549
1                16             17    363.000
```

```
[32]: winner = world_cup["Winner"].value_counts()
winner
```

```
[32]: Winner
Brazil      5
Italy       4
Germany FR  3
Uruguay     2
Argentina   2
England     1
France      1
Spain       1
Germany     1
Name: count, dtype: int64
```

```
[33]: runnerup = world_cup["Runners-Up"].value_counts()
runnerup
```

```
[33]: Runners-Up
Argentina      3
Germany FR     3
Netherlands    3
Czechoslovakia 2
Hungary        2
Brazil         2
Italy          2
Sweden         1
Germany        1
France         1
Name: count, dtype: int64
```

```
[34]: third = world_cup['Third'].value_counts()
third
```

```
[34]: Third
      Germany      3
      Brazil       2
      Sweden       2
      France       2
      Poland       2
      USA          1
      Austria      1
      Chile        1
      Portugal     1
      Germany FR   1
      Italy        1
      Croatia      1
      Turkey       1
      Netherlands  1
      Name: count, dtype: int64
```

```
[35]: teams = pd.concat([winner,runnerup,third],axis=1,keys=['Winner', 'Runner-up', '
      ↪Third'])
      teams.fillna(0,inplace=True)
      teams = teams.astype(int)
      teams
```

```
[35]:
```

	Winner	Runner-up	Third
Brazil	5	2	2
Italy	4	2	1
Germany FR	3	3	1
Uruguay	2	0	0
Argentina	2	3	0
England	1	0	0
France	1	1	2
Spain	1	0	0
Germany	1	1	3
Netherlands	0	3	1
Czechoslovakia	0	2	0
Hungary	0	2	0
Sweden	0	1	2
Poland	0	0	2
USA	0	0	1
Austria	0	0	1
Chile	0	0	1
Portugal	0	0	1
Croatia	0	0	1
Turkey	0	0	1

4 Attendance, Number of Teams, Goals, and Matches per Cup

```
[37]: world_cup.head(3)
```

```
[37]:   Year Country Winner Runners-Up Third Fourth GoalsScored \
0  1930  Uruguay  Uruguay   Argentina    USA  Yugoslavia      70
1  1934   Italy   Italy Czechoslovakia Germany   Austria      70
2  1938  France   Italy    Hungary   Brazil    Sweden      84

   QualifiedTeams  MatchesPlayed  Attendance
0              13              18    590.549
1              16              17    363.000
2              15              18    375.700
```

```
[38]: df=world_cup
```

```
[39]: df['Attendance'] = df['Attendance'].astype(str).str.replace('.', '').astype(int)
```

```
[40]: df.head(10)
```

```
[40]:   Year Country Winner Runners-Up Third Fourth \
0  1930  Uruguay  Uruguay   Argentina    USA  Yugoslavia
1  1934   Italy   Italy Czechoslovakia Germany   Austria
2  1938  France   Italy    Hungary   Brazil    Sweden
3  1950   Brazil  Uruguay    Brazil   Sweden    Spain
4  1954 Switzerland Germany FR   Hungary   Austria   Uruguay
5  1958   Sweden   Brazil    Sweden   France  Germany FR
6  1962   Chile   Brazil Czechoslovakia Chile  Yugoslavia
7  1966  England  England  Germany FR Portugal Soviet Union
8  1970   Mexico   Brazil    Italy  Germany FR   Uruguay
9  1974   Germany Germany FR Netherlands Poland    Brazil

   GoalsScored  QualifiedTeams  MatchesPlayed  Attendance
0           70              13              18    590549
1           70              16              17    363000
2           84              15              18    375700
3           88              13              22   1045246
4          140              16              26    768607
5          126              16              35    819810
6           89              16              32    893172
7           89              16              32   1563135
8           95              16              32   1603975
9           97              16              38   1865753
```

```
[122]: fig, axes = plt.subplots(2, 2, figsize=(20, 14))
```

```
# plot 1: Attendance per Year
```

```

sns.barplot(x=df['Year'].astype(str), y=df['Attendance'], ax=axes[0, 0],
            palette="viridis",width=0.5,edgecolor="black",linewidth=2)
axes[0, 0].set_title('Attendance per Year', fontsize=14)
axes[0, 0].set_xlabel('Year', fontsize=12)
axes[0, 0].set_ylabel('Attendance', fontsize=12)
axes[0, 0].tick_params(axis='x', rotation=45)

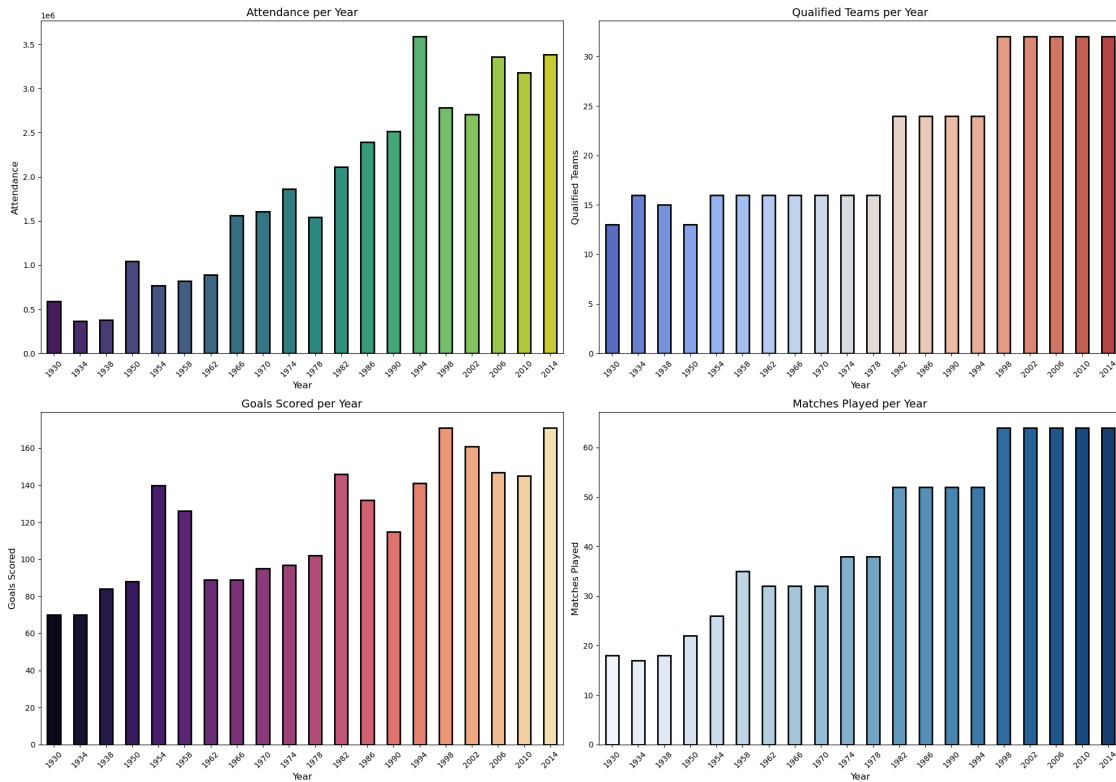
# plot 2: Qualified Teams per Year
sns.barplot(x=df['Year'].
            astype(str),y=df['QualifiedTeams'],ax=axes[0,1],palette='coolwarm',width=0.
            5,edgecolor="black",linewidth=2)
axes[0,1].set_title('Qualified Teams per Year',fontsize=14)
axes[0,1].set_xlabel('Year',fontsize=12)
axes[0,1].set_ylabel('Qualified Teams',fontsize=12)
axes[0,1].tick_params(axis='x',rotation=45)

# Plot 3: Goals Scored per Year
sns.barplot(x=df['Year'].
            astype(str),y=df['GoalsScored'],ax=axes[1,0],palette='magma',width=0.
            5,edgecolor="black",linewidth=2)
axes[1,0].set_title('Goals Scored per Year',fontsize=14)
axes[1,0].set_xlabel('Year',fontsize=12)
axes[1,0].set_ylabel('Goals Scored',fontsize=12)
axes[1,0].tick_params(axis='x',rotation=45)

# plot 4: Matches Played per Year
sns.barplot(x=df['Year'].
            astype(str),y=df['MatchesPlayed'],ax=axes[1,1],palette='Blues',width=0.
            5,edgecolor="black",linewidth=2)
axes[1,1].set_title('Matches Played per Year',fontsize=14)
axes[1,1].set_xlabel('Year',fontsize=12)
axes[1,1].set_ylabel('Matches Played',fontsize=12)
axes[1,1].tick_params(axis='x',rotation=45)

plt.tight_layout()
plt.show()

```



5 Trend of Goals per Match Over the Years

```
[43]: df.head(3)
```

```
[43]:   Year Country Winner Runners-Up Third Fourth GoalsScored \
0  1930  Uruguay  Uruguay   Argentina    USA  Yugoslavia         70
1  1934   Italy   Italy Czechoslovakia Germany   Austria         70
2  1938  France   Italy      Hungary   Brazil    Sweden         84

   QualifiedTeams  MatchesPlayed  Attendance
0              13             18    590549
1              16             17    363000
2              15             18    375700
```

```
[44]: df['Goals_per_Match'] = df['GoalsScored'] / df['MatchesPlayed']
```

```
[45]: df['Goals_per_Match']=df['Goals_per_Match'].round()
```

```
[46]: df['Goals_per_Match']=df['Goals_per_Match'].astype(int)
```

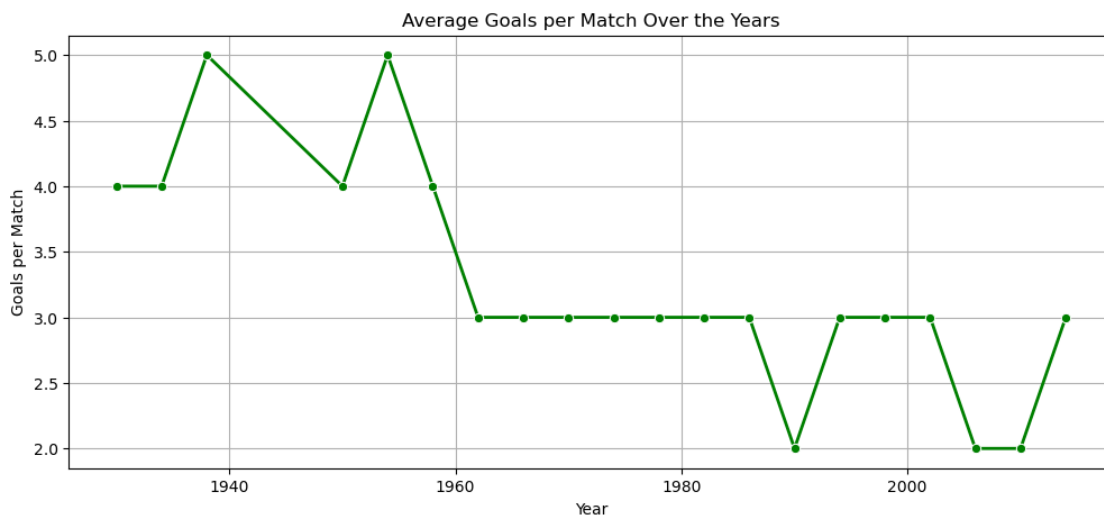
```
[47]: df.head(3)
```

```
[47]:
```

	Year	Country	Winner	Runners-Up	Third	Fourth	GoalsScored	\
0	1930	Uruguay	Uruguay	Argentina	USA	Yugoslavia	70	
1	1934	Italy	Italy	Czechoslovakia	Germany	Austria	70	
2	1938	France	Italy	Hungary	Brazil	Sweden	84	

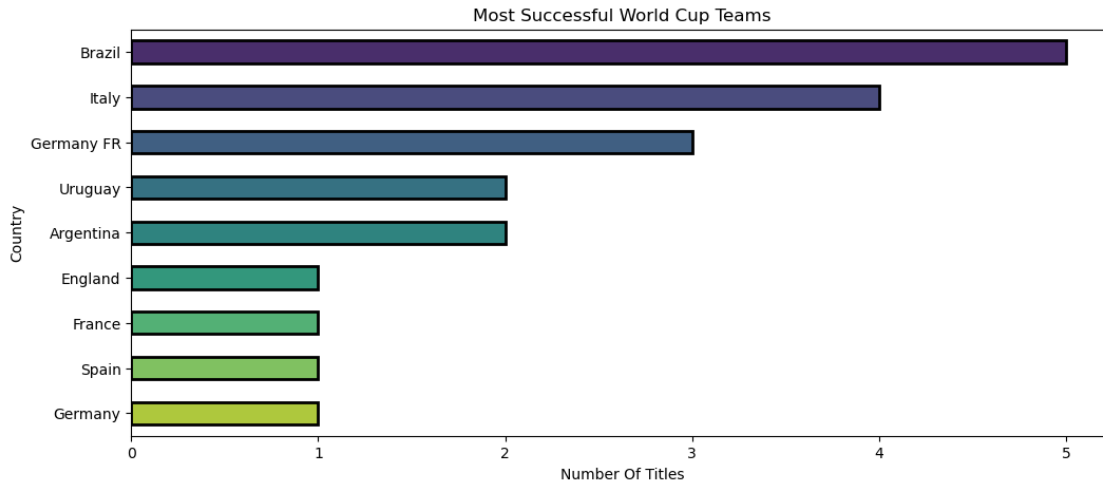
	QualifiedTeams	MatchesPlayed	Attendance	Goals_per_Match
0	13	18	590549	4
1	16	17	363000	4
2	15	18	375700	5

```
[124]: plt.figure(figsize=(12,5))
sns.
    ↳lineplot(x=df['Year'],y=df['Goals_per_Match'],marker='o',color='g',linewidth=2)
plt.title('Average Goals per Match Over the Years')
plt.xlabel('Year')
plt.ylabel('Goals per Match')
plt.grid(True)
plt.show()
```



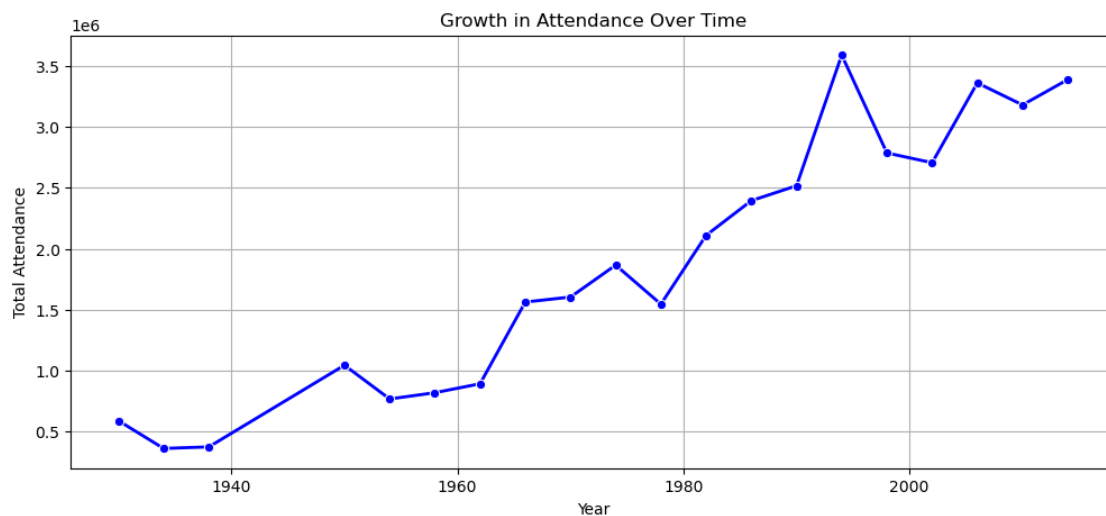
6 Top Winning Countries (Most Titles)

```
[126]: plt.figure(figsize=(12,5))
sns.countplot(y=df['Winner'],order=df['Winner'].value_counts().
    ↳index,palette='viridis',width=0.5,edgecolor="black",linewidth=2)
plt.title('Most Successful World Cup Teams')
plt.xlabel('Number Of Titles')
plt.ylabel('Country')
plt.show()
```



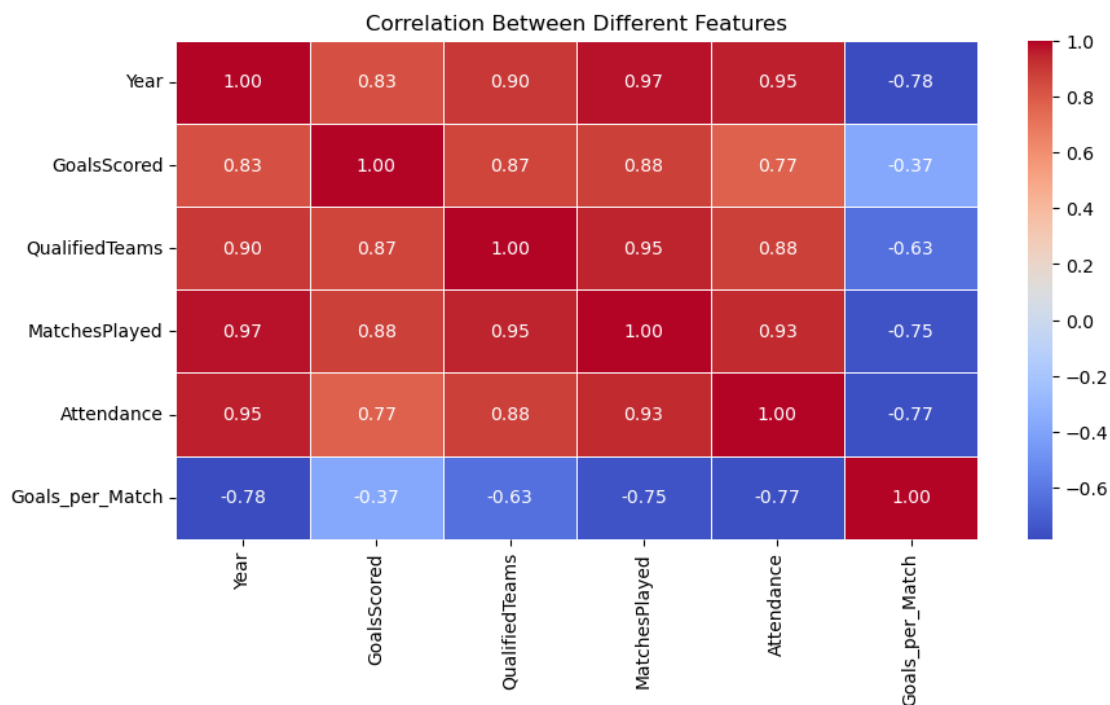
7 Growth in Attendance Over Time

```
[130]: plt.figure(figsize=(12, 5))
sns.lineplot(x=df['Year'], y=df['Attendance'], marker='o',
             color='b', linewidth=2)
plt.title('Growth in Attendance Over Time')
plt.xlabel('Year')
plt.ylabel('Total Attendance')
plt.grid(True)
plt.show()
```



8 Correlation Heatmap

```
[54]: plt.figure(figsize=(10,5))
sns.heatmap(df.corr(numeric_only=True),annot=True,cmap='coolwarm',fmt='.
    ↪2f',linewidths=0.5,)
plt.title('Correlation Between Different Features')
plt.show()
```



9 Relationship Between Goals & Matches Played

```
[56]: df.head(2)
```

```
[56]:   Year  Country  Winner  Runners-Up  Third  Fourth  GoalsScored  \
0  1930  Uruguay  Uruguay   Argentina    USA  Yugoslavia         70
1  1934   Italy   Italy  Czechoslovakia  Germany    Austria         70

   QualifiedTeams  MatchesPlayed  Attendance  Goals_per_Match
0              13             18     590549                4
1              16             17     363000                4
```

10 Number of Goals per Country (for winning teams)

```
[58]: df.head(6)
```

```
[58]:
```

	Year	Country	Winner	Runners-Up	Third	Fourth \
0	1930	Uruguay	Uruguay	Argentina	USA	Yugoslavia
1	1934	Italy	Italy	Czechoslovakia	Germany	Austria
2	1938	France	Italy	Hungary	Brazil	Sweden
3	1950	Brazil	Uruguay	Brazil	Sweden	Spain
4	1954	Switzerland	Germany FR	Hungary	Austria	Uruguay
5	1958	Sweden	Brazil	Sweden	France	Germany FR

	GoalsScored	QualifiedTeams	MatchesPlayed	Attendance	Goals_per_Match
0	70	13	18	590549	4
1	70	16	17	363000	4
2	84	15	18	375700	5
3	88	13	22	1045246	4
4	140	16	26	768607	5
5	126	16	35	819810	4

```
[59]: df_goals = df.groupby('Country', as_index=False)['GoalsScored'].sum()
df_goals.columns = ['Country', 'TotalGoals']
```

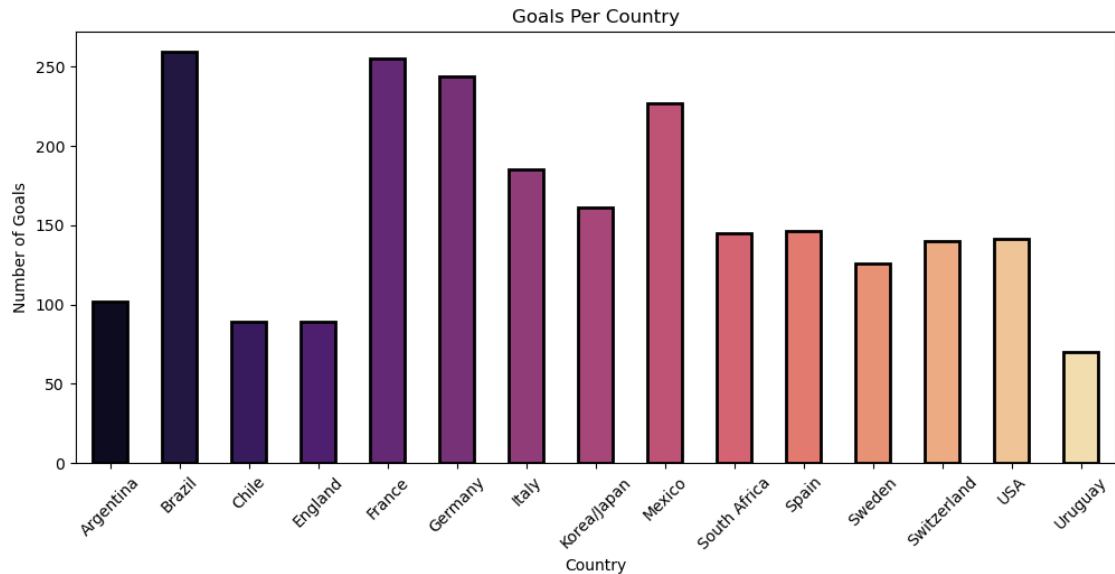
```
[60]: df_goals
```

```
[60]:
```

	Country	TotalGoals
0	Argentina	102
1	Brazil	259
2	Chile	89
3	England	89
4	France	255
5	Germany	244
6	Italy	185
7	Korea/Japan	161
8	Mexico	227
9	South Africa	145
10	Spain	146
11	Sweden	126
12	Switzerland	140
13	USA	141
14	Uruguay	70

```
[132]: plt.figure(figsize=(12,5))
sns.
    ↳ barplot(x=df_goals['Country'],y=df_goals['TotalGoals'],palette="magma",width=0.
    ↳ 5,edgecolor="black",linewidth=2)
plt.title('Goals Per Country')
```

```
plt.xlabel('Country')
plt.ylabel('Number of Goals')
plt.xticks(rotation=45)
plt.show()
```



11 Match outcome by home and away teams

```
[63]: def get_labels(matches):
        if matches['Home Team Goals'] > matches['Away Team Goals']:
            return 'Home Team Win'
        if matches['Home Team Goals'] < matches['Away Team Goals']:
            return 'Away Team Win'
        return 'DRAW'
```

```
[64]: matches['outcome'] = matches.apply(lambda x: get_labels(x), axis=1)
```

```
[65]: matches.head()
```

```
[65]:
```

	Year	Datetime	Stage	Stadium	City \
0	1930.0	13 Jul 1930 - 15:00	Group 1	Pocitos	Montevideo
1	1930.0	13 Jul 1930 - 15:00	Group 4	Parque Central	Montevideo
2	1930.0	14 Jul 1930 - 12:45	Group 2	Parque Central	Montevideo
3	1930.0	14 Jul 1930 - 14:50	Group 3	Pocitos	Montevideo
4	1930.0	15 Jul 1930 - 16:00	Group 1	Parque Central	Montevideo

	Home Team Name	Home Team Goals	Away Team Goals	Away Team Name \
0	France	4.0	1.0	Mexico

1	USA	3.0	0.0	Belgium
2	Yugoslavia	2.0	1.0	Brazil
3	Romania	3.0	1.0	Peru
4	Argentina	1.0	0.0	France

	Win conditions	...	Half-time Home Goals	Half-time Away Goals	\
0		...	3.0	0.0	
1		...	2.0	0.0	
2		...	2.0	0.0	
3		...	1.0	0.0	
4		...	0.0	0.0	

	Referee	Assistant 1	\
0	LOMBARDI Domingo (URU)	CRISTOPHE Henry (BEL)	
1	MACIAS Jose (ARG)	MATEUCCI Francisco (URU)	
2	TEJADA Anibal (URU)	VALLARINO Ricardo (URU)	
3	WARNKEN Alberto (CHI)	LANGENUS Jean (BEL)	
4	REGO Gilberto (BRA)	SAUCEDO Ulises (BOL)	

	Assistant 2	RoundID	MatchID	Home Team Initials	\
0	REGO Gilberto (BRA)	201.0	1096.0	FRA	
1	WARNKEN Alberto (CHI)	201.0	1090.0	USA	
2	BALWAY Thomas (FRA)	201.0	1093.0	YUG	
3	MATEUCCI Francisco (URU)	201.0	1098.0	ROU	
4	RADULESCU Constantin (ROU)	201.0	1085.0	ARG	

	Away Team Initials	outcome
0	MEX	Home Team Win
1	BEL	Home Team Win
2	BRA	Home Team Win
3	PER	Home Team Win
4	FRA	Home Team Win

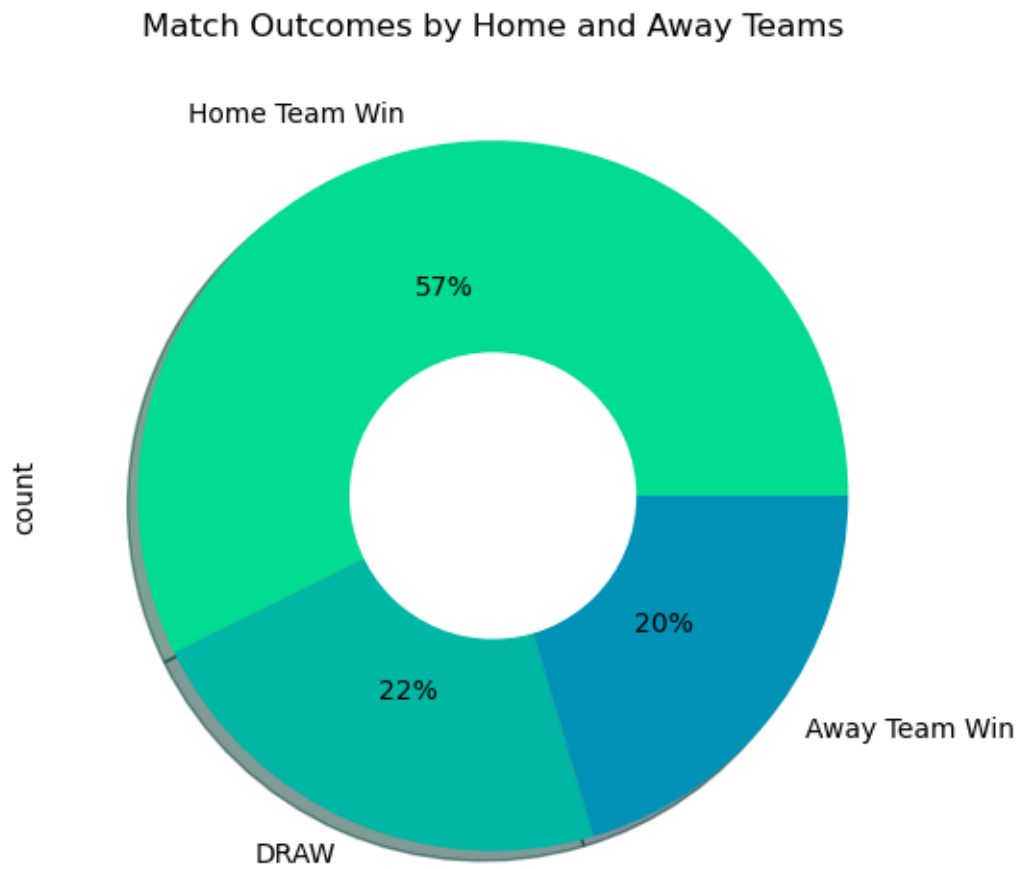
[5 rows x 21 columns]

```
[66]: mt = matches['outcome'].value_counts()
mt
```

```
[66]: outcome
Home Team Win    488
DRAW             190
Away Team Win    174
Name: count, dtype: int64
```

```
[67]: plt.figure(figsize=(6,6))
mt.plot.pieautopct='%1.0f%%',colors=sns.color_palette('winter_r'),shadow=True)
c=plt.Circle((0,0),0.4,color='white')
```

```
plt.gca().add_artist(c)
plt.title('Match Outcomes by Home and Away Teams')
plt.show()
```



[]:

[]: